University of Pretoria

Semester test: 17 May 2025

Course: Computer Science
Paper: COS332

Time: 90 minutes
Marks: 50

Examiner: Prof MS Olivier

This paper consists of 11 pages.

Answer all questions.
Casio FX81 or equivalent calculators are permitted, as well as simple 5-function calculators
If a question can be answered using an acronym only the the acronym is required.
You are allowed to write on this test paper.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) If MIME is deemed to be a data communications protocol it would fit best on layer … of the ISO OSI model.

   A: 7
   B: 6
   C: 5
   D: 4
   E: 3

2) Which character set is assumed to be understood by all parties involved in an HTTP exchange?

   A: ASCII
   B: EBCDIC
   C: Unicode
   D: UTF-8
   E: ISO-8859-1

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3) Consider the following protocol negotiation string in an HTTP request:
   `Accept-Charset: iso-8859-1, utf-8, utf-16, *;q=0.1`
   Suppose the server supports `iso-8859-1`, `utf-8` and `Shift_JIS`. The response may then be encoded using

   A: `iso-8859-1`
   B: `utf-8`
   C: `Shift_JIS`
   D: Any of the encodings listed as options above.
   E: One of two of the encodings listed as options above.

4) Which of the following is *not* a valid MIME type?

   A: binary
   B: example
   C: image
   D: message
   E: model

5) Suppose a message consists of JPG images embedded in HTML text. Which MIME type and subtype will be used to describe the message?

   A: `text/plain`
   B: `text/html`
   C: `image/jpeg`
   D: `multipart/mixed`
   E: `multipart/alternative`

6) Suppose an email consists of a message encoded in HTML and a JPG image (both included in the message). These two items are the only components of the email message. Which MIME type will be used to describe the *entire* email message?

A: `application/email`

B: `application/rfc822`

C: `multipart/alternative`

D: `multipart/mixed` ✓

E: `text/html`

7) How many Unicode characters can, in principle, be represented in UTF-8 using exactly four bytes? If the answer is expressed in the form 2^n, what is the value of n?

A: 14

B: 20

C: 21 ✓

D: 22

E: 27

8) Consider the byte sequence `D7 E5`. The Unicode character represented by this UTF-8 encoding is a(n)

A: Accented character.

B: Greek character.

C: Emoji.

D: Punctuation mark.

E: `D7 E5` is not a valid UTF-8 encoding. ✓

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9) A BER value in ASN.1 is, in principle, encoded as a triple consisting of

A: A type, a subtype and a value.

B: A type, a length and a value. ✓

C: A constructor and two operands.

D: A variable name, as well as its minimum and maximum values.

E: The same value encoded in binary, text and hexadecimal.

\#10) Which of the following is not a valid data type in ASN.1?

A: STRUCT ✗

B: CHOICE ✓

C: SEQUENCE ✓

D: ARRAY ✓

E: INTEGER ✓

Alternative is other

11) The syntactic structure of an ASN.1 message is defined using

A: A grammar ✓

B: An informal description in a natural language

C: A diagrammatic depiction of the message

D: A bit pattern

E: A list of the types of the components of the message

12) The OSI layer that, amongst others, attempts to emulate database transactions where messages can be 'rolled back' if the final message in a sequence is not sent, is layer

A: 2

B: 3 ✗

C: 4

D: 5 ✗

E: 6 ✗

11110

13) Which of the following data streams generally work(s) better with an unreliable layer 4 protocol?

A: Web page ?
B: File transfer
C: Audio
D: More than one of the above
E: All of the above

14) Which of the following protocol(s) is/are used on the transport layer by DNS?

A: TCP
B: UDP
C: IP
D: More than one of the above
E: All of the above

15) Which command enables one to see the status of transport layer connections on a host (in most operating systems)?

A: `netstat`
B: `ping`
C: `tracert` / `traceroute`
D: `tcp-show`
E: `ps`

16) UDP uses the following flow-control mechanism:

A: Stop and Wait
B: Sliding window
C: Unrestricted
D: More than one of the above
E: All of the above

17) A sends a TCP segment to node B. The acknowledgement field contains the value 150. The ACK flag is not set. This means

A: B may assume that bytes up to byte 149 that it had sent are acknowledged.
B: B should disregard the value 150 — it has no meaning.
C: This is a negative acknowledgement of the segment B has sent that had the sequence number 150.
D: The TCP layer at A is misconfigured; it should have set the ACK flag.
E: More than one of the above

18) Which of the following fields does *not* occur in a TCP header?

A: Source port
B: Window advertisement
C: Sequence number
D: Options
E: Length —> Data offset

19) Which of the following statements is/are *false* about phantom bytes used in an initial 3-way TCP/IP handshake?

A: It is counted as part of the SYN message.
B: It is counted as part of the SYN+ACK message.
C: It is counted as part of the ACK message.
D: It consists of one byte of data.
E: More than one of the statements above are false.

20) A TCP connection in the `TIME-WAIT` state means that

A: The node is waiting for data from the node that it is connected to.

B: The connection is about to time-out if no further data is transmitted soon.

C: The connection will soon be established (as soon as the handshake is completed).

D: The node is waiting for 'lost' traffic to arrive at the port that is no longer in use, before it will be available for reuse. → Quiet?

E: The network is congested.

21) A TCP node that is in the `LISTEN` state

A: Is acting as a server and waiting for a SYN message.

B: Is acting as a client and has sent a SYN message.

C: May act as a client after sending a SYN message.

D: More than one of the above

E: None of the above.

22) What does the claim that QUIC is a *quick* protocol mean?

A: It manages the lower layers to transmit raw data at higher bit rates

B: It reduces latency when establishing or re-establishing a connection.

C: Where multiple parts of a message has to be transported it ensures that parts of the message are delivered quickly even if some parts are delayed.

D: More than one of the above

E: All of the above

23) QUIC connections can migrate. Which party can initiate such a migration in QUICv1 or QUICv2?

A: The client

B: The server

C: The network management system

D: More than one of the above

E: All of the above

24) The RIR for Europe is

A: RIPE

B: EurNIC

C: EuroNIC

D: EurIN

E: EPIR

25) If one sends a message to an IP multicast address, the message will be delivered to

A: The node to which that IP address has been assigned.

B: All nodes on the local network.

C: All nodes that are members of some group.

D: The network itself, rather than to a host on the network.

E: To the network gateway that connects the network to other networks. **[25]**

Question 2

All values is this question are expressed in hexadecimal notation.

- Convert the following *Unicode* characters to byte sequences. Represent byte sequences as 2-digit hexadecimal numbers with a space between all bytes. (Your answers will therefore be similar to a string such as `12 bf 3a`.)

  ✓ a) U+1842
  ✓ b) U+10AD0
  ✓ c) U+2CA0

- UTF-8 encoding is self-synchronising. Consider the following byte sequence, which is not a valid UTF encoding:

  `47 CE B5 C4 85 E1 80 AA E1 9C A0`
  `[illegible] AD F0 90 8F 8C 57 73 E1 83 A8`

  d) Provide the first valid Unicode character that occurs in this sequence. Provide an answer in the form U+xxxx.

  e) Provide the second valid Unicode character that occurs in this sequence. Provide an answer in the form U+xxxx.

  f) Provide the subsequence of bytes that are invalid.

  g) Convert the bytes starting at the point where synchronisation is re-established, convert those that form the first Unicode character, and provide that character in the typical format (U+xxxx) on your answer sheet.

  h) Provide the last valid Unicode character that occurs in this sequence. Provide an answer in the form U+xxxx.

- We did not discuss UTF-16 and UTF-32 at all. It should be obvious that UTF-16 represents Unicode characters as sequences of 16-bit words, while UTF-32 represents Unicode characters as sequences of 32-bit words. When representing words as bytes, the endianness of the representation matter. Recall that big-endian representation uses the 'natural' sequence of bytes. Hence the word 1234 is represented as the bytes `12 34`, and the word 12345678 is represented as the bytes `12 34 56 78`. Use big-endian representations for the questions that follow. Use the principles of UTF-8 to determine the following representations. (Unfortunately, the principles of UTF-8 do not generalise well for cases that look different from what follows.)

  ☆ i) Represent `U+0048` in UTF-16 as a byte sequence.
  ☆ j) Represent `U+0F40` in UTF-32 as a byte sequence. **[10]**

Question 3

Very briefly state what happens when the TCP timers listed below expire. Keep your answer brief. For example, if expiration of some timer means that a segment containing data would be transmitted, simply say *A segment containing data is transmitted.* There is no need to explain the nature of the data, the reason why the data is transmitted or provide any other details. (Having said this, *A segment containing data is transmitted* would not be a correct answer for any of the timers listed below.)

Note that the space to provide your answers start on the second page of the answer sheet.

a) Retransmission timer – retransmit

b) Acknowledgement timer

☆ c) Persistence timer – send probe to check still there

d) Keepalive timer – connection idle

e) Quiet timer – wait for delayed segments **[5]**

Question 4

TCP is a reliable protocol. List any five mechanisms used by TCP to provide this reliability. Keep your answers brief; no explanations are required. **[5]**

Flow-control { Stop-wait, Window-slider

Connections

Timers

ARQ

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Question 5

Answer the following questions about IPv4 addresses. Use CIDR notation for all your answers.

a) Which IPv4 address is used in a routing table to indicate that the routing table entry points to the default gateway?

b) A block of 256 'class C' network addresses is reserved for private use. Provide the *first* class C network address that occurs within this block.

c) A block of 16 'class B' network addresses is reserved for private use. Provide the *last* class B network address that occurs within this block.

d) 137.215.98.140 is a host on the UP network. What is the UP network address?

e) 8.8.8.8 is a popular DNS server provided by Google. If this address were still part of a class-based address, what would the broadcast address on this network be?

Please recall that CIDR notation has to be used for all your answers in this question. **[5]**

TOTAL **[50]**

END OF PAPER

0 - 127....
128 - 191...
192 - 223...
224 - 240...
241 - ...

A : 0xxxxxxx   1n 3h
B : 10xxxx xxxxxxxx ~~2n2h~~ 2N 2H
C : 110xxxxx xxxxxxxx xxxxxxxx 3N 1H
D : 1110....
E : 1111